

KNN Error Convergence - Beginner Explanation

Step 1: Recall what KNN does

- KNN (K-Nearest Neighbors) is like asking your friends for advice.
- If you want to decide something, you look at your K nearest friends (neighbors) and go with what most of them say.

Step 2: Error in KNN

- Error means: how many times KNN gives a wrong answer.
- At the start (with little data), the error might be high because KNN doesn't know much.

Step 3: What is "Error Convergence"?

- Convergence means: "settling down to a stable value".
- In KNN, when we add more and more data, the error rate keeps going down (we make fewer mistakes).
- But after a certain point, the error does not decrease much anymore — it converges (stops changing much).

Analogy: If you've only seen 2 apples and 1 banana, you might make mistakes.

If you've seen 100 apples and 100 bananas, you'll make fewer mistakes.

If you've seen 10,000 apples and bananas, adding more won't help much — you already recognize fruits almost perfectly.

This "almost perfect but not 0" is the error convergence.

Mathematical Definition of Error Convergence

- Error rate:
 $P(\text{error}) = P(\hat{Y} \neq Y)$
- Bayes error (best possible error):
 $P^* = P(\text{Bayes error})$
- Error convergence formula:
 $\lim_{n \rightarrow \infty} P(\text{KNN error}) \rightarrow P^*$

Numerical Example of Error Rate

Suppose 10 test fruits with true and predicted labels.

Out of 10, 2 predictions are wrong.

Error rate = $2/10 = 20\%$

Bayes Error Calculation

Bayes classifier chooses the class with the highest probability:

$$\hat{Y}(x) = \operatorname{argmax}_c P(Y=c | X=x)$$

Bayes error:

$$P^* = E_X [1 - \max_c P(Y=c | X)]$$

Example:

If $P(\text{Apple}|x) = 0.7$ and $P(\text{Banana}|x) = 0.3$,

Bayes classifier picks Apple.

Error = $1 - 0.7 = 0.3$

Two-Class Toy Example

Input types: Green (G) and Yellow (Y)

Probabilities:

- Green: $P(\text{Apple}|G)=0.6$, $P(\text{Banana}|G)=0.4$

- Yellow: $P(\text{Apple}|Y)=0.2$, $P(\text{Banana}|Y)=0.8$

Each equally likely.

Bayes error = $(0.5 \cdot 0.4 + 0.5 \cdot 0.2) = 0.3$ (30%)

Relation Between 1-NN and Bayes Error

Theorem: $P(1NN) \leq 2P^*$

This means the 1-NN classifier is at most a factor 2 worse than the Bayes classifier.

Numerical Verification Example:

Green: $P(\text{Apple})=0.7$, $P(\text{Banana})=0.3$

Yellow: $P(\text{Apple})=0.4$, $P(\text{Banana})=0.6$

Bayes error = 0.35

1-NN error = 0.45

Check: $0.45 \leq 2 \cdot 0.35 = 0.7$ (True)

Reason Behind the Factor of 2

At each point x :

- Bayes error: $P^*(x) = \min(p, 1-p)$

- 1-NN error: $P(1NN)(x) = 2p(1-p)$

For both cases ($p \leq 0.5$ or $p \geq 0.5$), it can be shown that

$P(1NN)(x) \leq 2P^*(x)$

Therefore overall:

$P(1NN) \leq 2P^*$

Conclusion:

- KNN error decreases with more data and converges to Bayes error.
- Bayes error is the theoretical lower bound of error due to uncertainty in data.
- 1-NN is guaranteed to be at most 2 times worse than Bayes error.